Fundamental steps in digital image processing

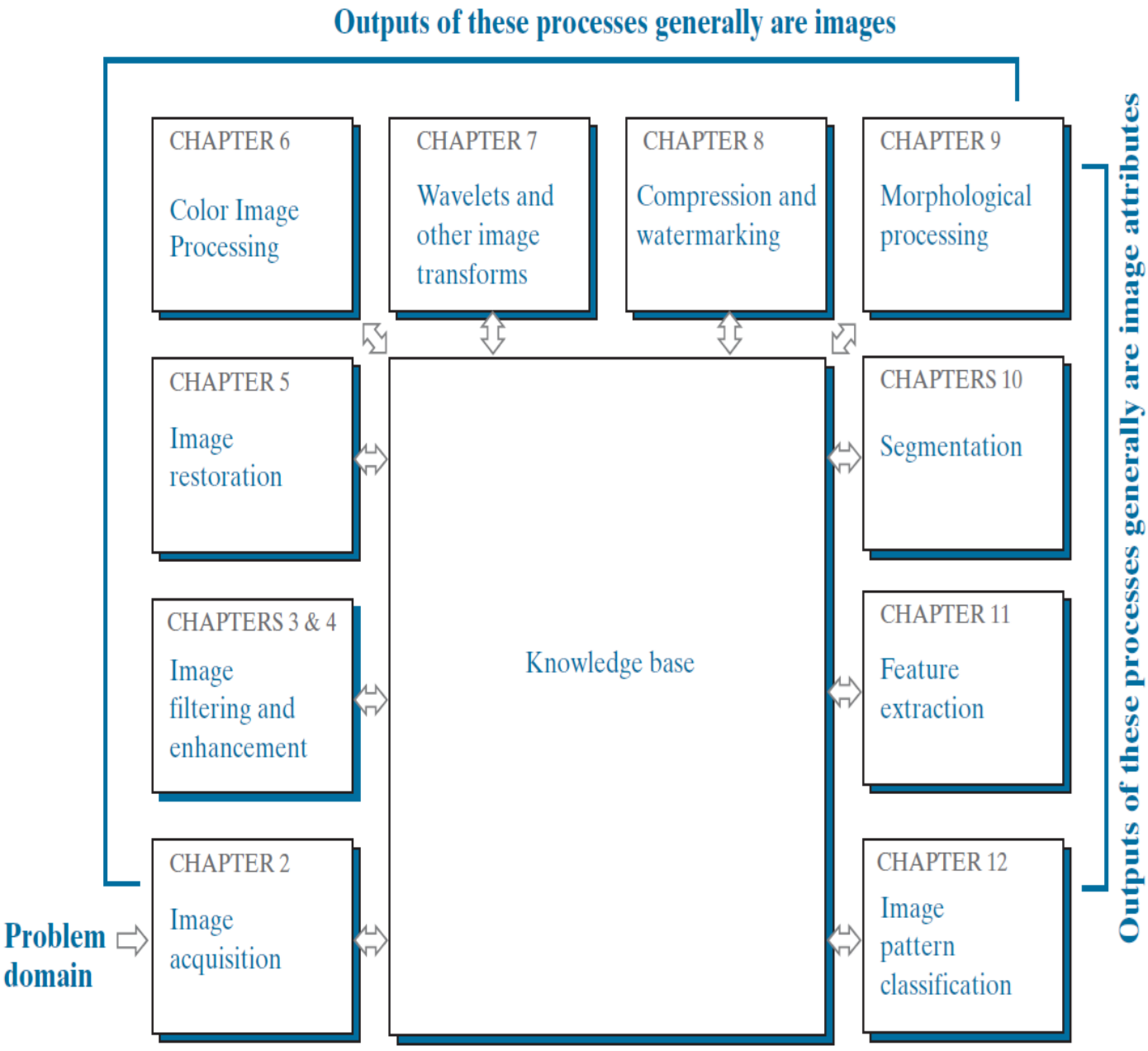


FIGURE 1.23 page 41

General-purpose image processing system.

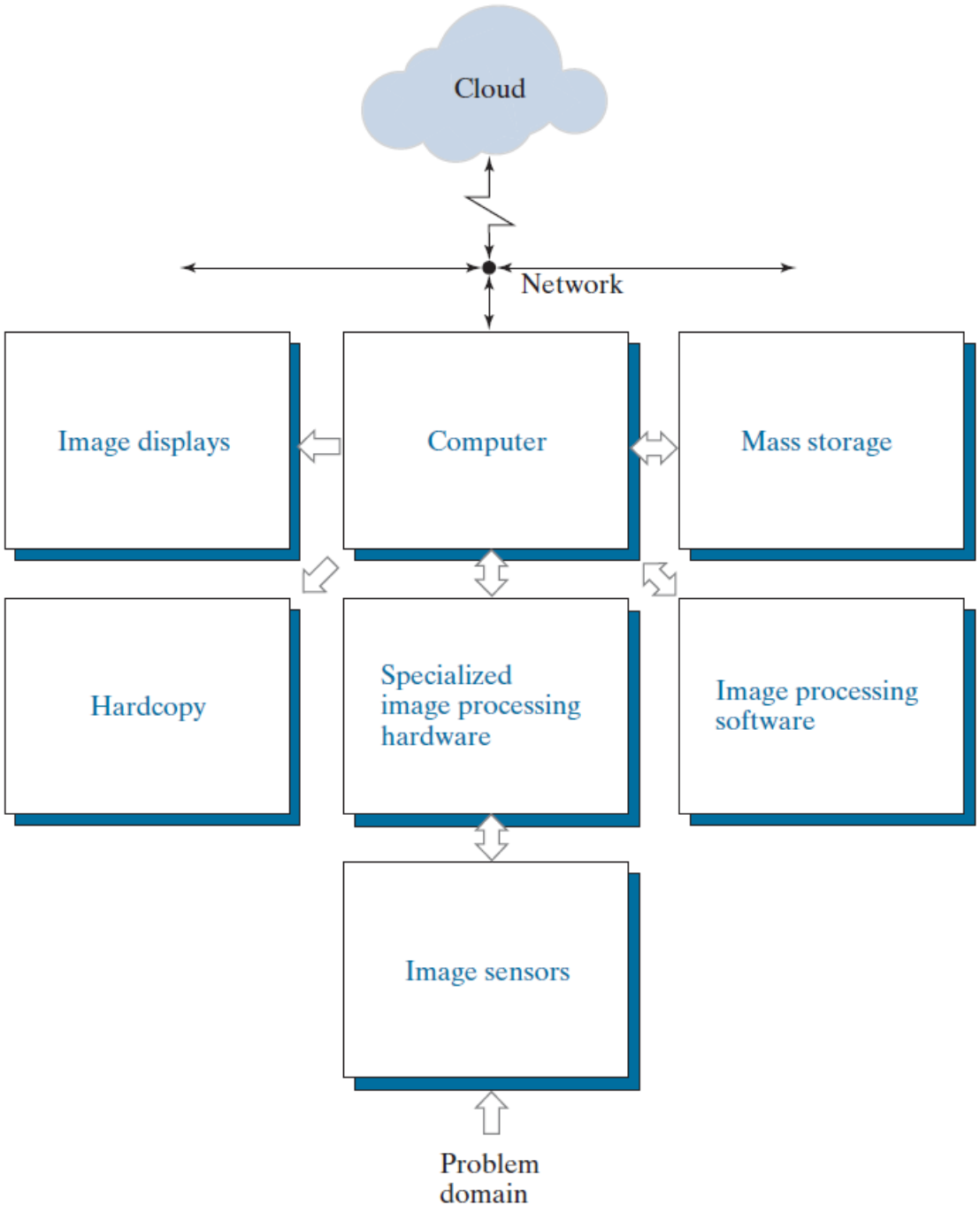
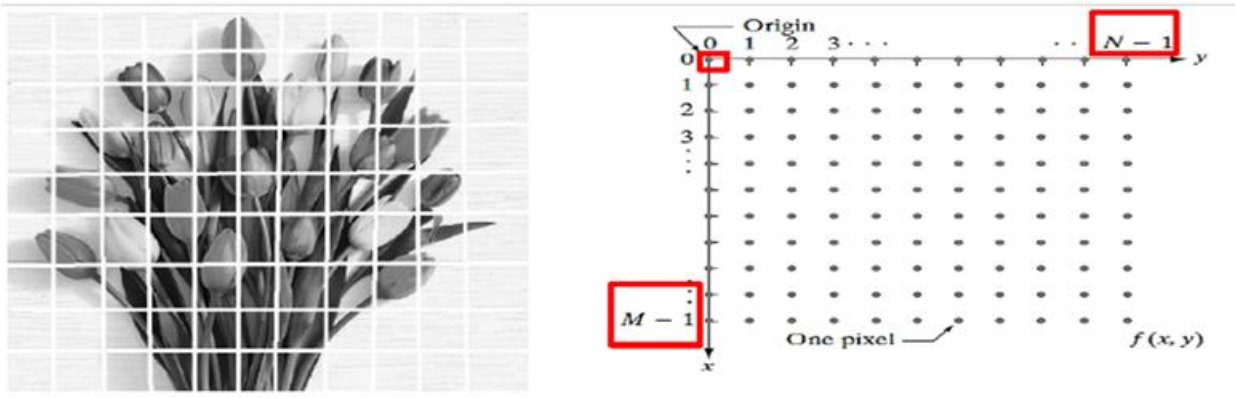


FIGURE 1.24 page 45

CMPn332 – CMPN332 – CMP432

Digital Image



$$F = \begin{bmatrix} 52 & 55 & 61 & 66 & 70 & 61 & 64 & 73 \\ 63 & 59 & 55 & 90 & 109 & 85 & 69 & 72 \\ 62 & 59 & 68 & 113 & 144 & 104 & 66 & 73 \\ 63 & 58 & 71 & 122 & 154 & 106 & 70 & 69 \\ 67 & 61 & 68 & 104 & 126 & 88 & 68 & 70 \\ 79 & 65 & 60 & 70 & 77 & 68 & 58 & 75 \\ 85 & 71 & 64 & 59 & 55 & 61 & 65 & 83 \\ 87 & 79 & 69 & 68 & 65 & 76 & 78 & 94 \end{bmatrix}$$

F(2,1)

Access image information and pixels values

```
#include<opencv2/opencv.hpp>
#include"iostream"
using namespace std;
using namespace cv;
int main()
{
	Mat src3 = imread("x.jpg",0);
	cout << "pixels intensities = \n"<< src3<< endl;
	cout << "image rows=" << src3.rows << endl;
	cout << "image cols=" << src3.cols << endl;
	cout << "image pixel total=" << src3.total() << endl;
	cout << "image channels=" << src3.channels() << endl;
	cout << "image pixel depth=" << src3.depth() << endl;
	                              //return a number from 0 to 6
	int pixel_v = src3.at<uchar>(0, 0);
	cout << "pixel value(0,0)=" << pixel_v << endl;
	namedWindow("img", 0);
	imshow("img", src3);
	imwrite("img.Jpg", src3);
	waitKey(0);


}
```

Output

